ARANYA SAHA

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EDUCATION

University of Maryland, College Park (UMD) Aug 2026 – Present
Ph.D. in Electrical and Computer Engineering
Research Advisor: Prof. Sanghamitra Dutta

Bangladesh University of Engineering and Technology (BUET) Feb 2020 – Mar 2025
B.Sc. in Electrical and Electronic Engineering
Research Advisor: Prof. Mohammad Ariful Haque

RESEARCH INTERESTS

Reinforcement Learning for LLMs, LLM Unlearning, Mechanistic Interpretability, Information Theory.

ACADEMIC AND PROFESSIONAL EXPERIENCE

University of Maryland, College Park (UMD) Aug 2026 – Present
Graduate Research Assistant *Maryland, USA*

Advanced Chemical Industries PLC Apr 2025 – Aug 2026
Machine Learning Engineer *Dhaka, Bangladesh*

Bangladesh University of Engineering and Technology (BUET) Nov 2025 – Apr 2026
Part-time Lecturer, Department of EEE *Dhaka, Bangladesh*

Institute of Robotics and Automation, BUET Jun 2025
Instructor, Robotics Bootcamp 2025 *Dhaka, Bangladesh*

PUBLICATIONS

Preprints

1. Ismam Nur Swapnil, **Aranya Saha**, Tasneea Zahra, Tanvir Ahmed Khan, Mohammad Ariful Haque, Ser-Nam Lim, "Skip What You Can Predict: Predictive Repositioning for Policy Optimization for Efficient LLM Training." [arXiv]
2. **Aranya Saha***, Tanvir Ahmed Khan*, Ismam Nur Swapnil*, Mohammad Ariful Haque, "CLARIFY: A Specialist–Generalist Framework for Accurate and Lightweight Dermatological Visual Question Answering." [arXiv]
3. Ismam Nur Swapnil*, **Aranya Saha***, Tanvir Ahmed Khan*, Mohammad Ariful Haque, "GRPO++: Enhancing Dermatological Reasoning under Low Resource Settings." [arXiv]
4. Tanvir Ahmed Khan, **Aranya Saha**, Ismam Nur Swapnil, Mohammad Ariful Haque, "Compression Strategies for Efficient Multimodal LLMs in Medical Contexts." [arXiv]

Conference Publications

1. Shadman Sobhan, **Aranya Saha**, Tanvir Ahmed Khan, Abduz Zami, "Skin Cancer Classification Using Pre-trained CNNs: A Transfer Learning Approach Addressing Imbalanced Data Challenges," International Conference on Next-Generation Computing, IoT and Machine Learning, 2025. [IEEE Xplore]
2. Shadman Sobhan, Abduz Zami, Mohiuddin Ahmed, Tanvir Mahtab Zihan, Tanvir Ahmed Khan, **Aranya Saha**, "A Multi-Stage Deep Learning Approach to Tuberculosis Detection with Explainable Insights," International Conference on Next-Generation Computing, IoT and Machine Learning, 2025. [IEEE Xplore]

SELECTED PROJECTS

1. **Efficient Frame Selection for Long Egocentric Video Understanding.**
   Developed an efficient frame selection approach for long-form egocentric video understanding. [GitHub]
2. **EchoLens: Multimodal Conversational AI Engine.**
   Developed a multimodal conversational system integrating vision and language models. [GitHub]
3. **Simple MedQA-GPT: Domain-Specific Medical Question Answering.**
   Developed a GPT-based medical question-answering system using domain-specific prompting. [GitHub]
4. **IoT-Based Patient Health Monitoring System.**
   Designed an IoT-enabled system for real-time patient health monitoring. [GitHub]
5. **MATLAB-Based Fingerprint Recognition System.**
   Implemented a fingerprint recognition system using digital image processing techniques. [GitHub]

RELEVANT COURSEWORK

Graduate	Random Processes, Advanced Digital Signal Processing
Undergraduate	Artificial Intelligence and Machine Learning, Digital Signal Processing, Random Signals and Processes, Probability and Statistics, Signals and Systems

TECHNICAL SKILLS

Programming	Python, C/C++, MATLAB
ML/DL	PyTorch, TensorFlow, CNNs, VLMs, LLMs
Tools	NumPy, Pandas, Git, Docker, FastAPI, LaTeX
Hardware	Microcontrollers, IoT Devices, Sensors

LEADERSHIP EXPERIENCE

Association for Computing Machinery (ACM) – SIGCOMM Apr 2024 – Feb 2025
Student Executive *Remote*

Served as the inaugural Student Executive of ACM SIGCOMM, working with Prof. Matthew Caesar. Developed the organization's official website and co-established a paper reading group.

HONORS AND AWARDS

- Dean's Fellowship, Department of ECE, University of Maryland (2026–2027).
- 1st Runner-Up, AI Poster Competition, BEAR Summit 2025.
- University Merit Scholarship, BUET (4 semesters).
- Dean's List Award, BUET (Years 1–2).
- Ranked 29th among 10,000+ candidates in the BUET Undergraduate Admission Test (2019).
- Ranked 31st among male candidates in the Dhaka Board HSC Examination; recipient of the Talent Pool Scholarship.

REFERENCES

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